

A 2-approximation algorithm for the zookeeper's problem

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Abstract

Let P be a simple polygon, and let $\mathcal{P}$ be a set of disjoint convex polygons inside P , each sharing one edge with P . The *zookeeper's route problem* asks for a shortest route inside P that visits (but does not enter) each polygon in $\mathcal{P}$. We present an $O(n)$ time algorithm for computing a zookeeper's route of length at most 2 times that of the shortest zookeeper's route. Our result improves upon the previous approximation factor 6.

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1. Introduction

Shortest paths are of fundamental importance in robotics and computational geometry. In this paper, we study the following *zookeeper's problem* [4]. Given a simple polygon P (the *zoo*) with a set $\mathcal{P}$ of disjoint convex polygons (the *cages*) inside it, each sharing one edge with P , find a shortest route inside P that visits (without entering) each polygon in $\mathcal{P}$. One may consider it as minimizing the route for a zookeeper to feed animals.

Let n denote the total number of edges of the polygon P and polygons in $\mathcal{P}$. An $O(n^2)$ time algorithm for computing a shortest zookeeper's route has been presented by Tan [8]. Apart from the general zookeeper's problem described above, there is a restricted version of the problem in which a starting point s on the boundary of P is given. Chin and Ntafos were the first to

give an $O(n^2)$ time algorithm for computing the shortest *fixed* zookeeper's route, i.e., the route is forced to pass through the point s [4]. This result was later improved to $O(n \log^2 n)$ by Hershberger and Snoeyink [6], and recently to $O(n \log n)$ by Bepamyatnikh [1]. Most of these exact algorithms make use of the unfolding technique, which unfolds a polygon or the polygon triangulation by reflecting it on some line segments in the polygon $O(n)$ times. They are not only complicated but also far from practical. If the input coordinates have L bits of precision, then the output has $O(nL)$ bits [6]. Hence, these exact algorithms suffer from numerical problems.

A practical approach to the zookeeper's problem is to develop simple, approximative solutions. Such a work has been done by Jonsson, who gave an $O(n)$ time algorithm for computing a zookeeper's route of length at most 6 times that of the shortest zookeeper's route [7]. For the restricted problem in which a point s on the boundary of P is given, Tan has also developed an $O(n)$ time algorithm for computing a zookeeper's route that

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passes through s and is of length at most $\sqrt{2}$ times that of the shortest fixed zookeeper's route [11]. Both approximation algorithms are not only faster than the best exact algorithms, but also vastly simpler because they require only the computation of shortest paths inside the interior of the zoo (i.e., inside $P - \mathcal{P}$).

In this paper, we present an $O(n)$ time algorithm for computing a zookeeper's route of length at most 2 times that of the shortest zookeeper's route. We first select a point s on some edge of a cage, and then compute the zookeeper's route through s using the known $\sqrt{2}$ -approximation algorithm [11]. The point s is chosen so that neither the incoming angle nor the outgoing angle of the shortest zookeeper's route through s with the edge containing s can exceed $\pi/2$. This enables us to obtain the approximation factor 2. Our result improves upon the previous approximation factor, and the techniques we employ in this paper can probably be used for approximating other kinds of routes as well.

2. Known solutions to the zookeeper's problem

Let us first give a brief review of Chin and Ntafos' algorithm for computing the shortest fixed zookeeper's route [4]. Suppose that we have a set S of edges, one per cage. See Fig. 1 for an example, where the edges of S are drawn in bold lines. The locally optimal zookeeper's route, which makes contacts with the edges of S ,

can be computed as follows. First, triangulate the interior of $P - \mathcal{P}$ (Fig. 1(a)). Next, unfold the triangulation using the edges of S as mirrors, in their order on the boundary of P , and find the shortest path between s and its image s' in the unfolded polygon (Fig. 1(b)). Finally, fold back the shortest path to obtain the locally optimal zookeeper's route (Fig. 1(c)). This process takes linear time, since a simple polygon can be triangulated in linear time [2] and the shortest path between two points in a polygon can be found in linear time [5]. Particularly, the edges of S is called the *active edges* for the locally optimal route computed above, and the found route is said to *reflect* on active edges.

Chin and Ntafos' algorithm first selects an initial set S_0 of edges, one per cage, and computes the initial zookeeper's route R_0 using the above unfolding method. The shortest fixed zookeeper's route R is then found by repeatedly adjusting the current route R_i until it becomes the shortest one. Let e_x and e_y denote two adjacent edges of a cage $P_j \in \mathcal{P}$. The route R_i is *adjustable* on the active edge e_x if the contact of R_i with e_x is only the common point of e_x and e_y , and the incoming (resp. outgoing) angle formed by R_i with the line containing e_y is smaller than the outgoing (resp. incoming) angle formed by R_i with the line containing e_y . Fig. 2 shows two different cases in which an adjustment occurs. The *direction of an adjustment on a cage* is defined as the shift direction of two reflection points on the cage, before and after the adjustment is performed.

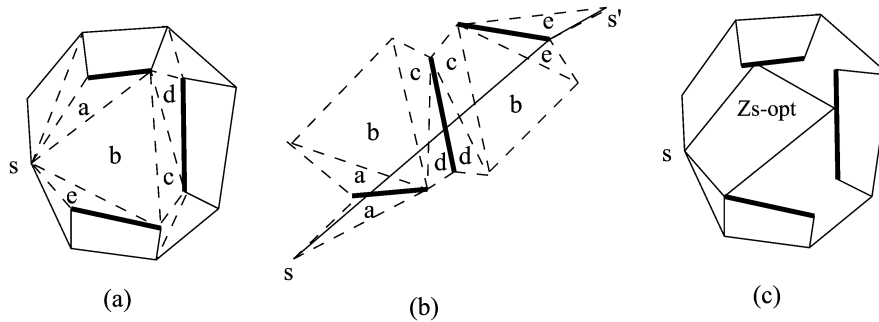


Fig. 1. Construction of a zookeeper's route.

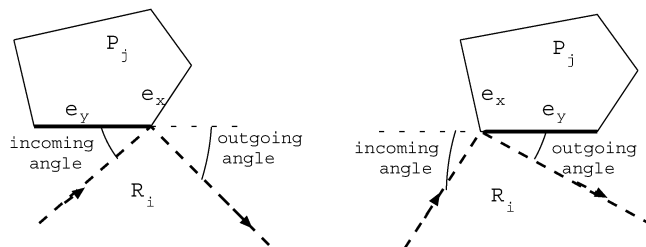


Fig. 2. Illustration of adjustments.

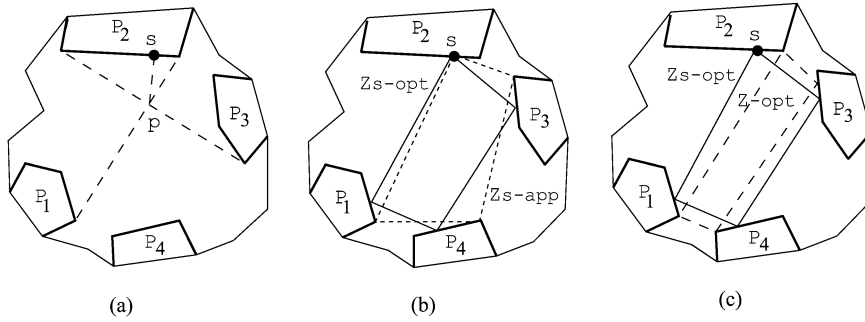


Fig. 3. Illustration for the point s and the routes Z_{s-opt} , Z_{s-app} and Z_{opt} .

An adjustment involves a change in the set of active edges (i.e., e_y is substituted for e_x in the new set S_{i+1}) and thus calls the unfolding procedure once to compute the new and shorter zookeeper's route. When the current route cannot be adjusted any more, it gives the shortest fixed zookeeper's route. See [4,6] for details.

A quadratic time algorithm for removing the starting point constraint has been presented in [8]. Before describing it, we first show the uniqueness of the shortest zookeeper's route.

Lemma 1. *There is a unique non-adjustable zookeeper's route in the interior of the zoo (i.e., in $P - \mathcal{P}$), except for very special cases where there is an infinite number of non-adjustable routes of equal length.*

Proof. The lemma can be proved by an argument similar to that given in the proof of [9, Lemma 4]. $\square$

Lemma 2. *A zookeeper's route is a shortest zookeeper's route if and only if it is non-adjustable.*

Proof. The lemma can be proved by an argument similar to that given in the proof of [3, Corollary 1]. $\square$

The starting point constraint can then be removed as follows. First, take a point s on same cage of $\mathcal{P}$ arbitrarily, and compute the shortest zookeeper's route through s . Then, slide the reflection point of the current zookeeper's route, in the direction of possible adjustments, along the boundary of the chosen cage. During the sliding process, we maintain the shortest zookeeper's routes through the so-called *event points*, where either the unfolded polygon is changed or a shortest zookeeper's route is obtained. It follows from Lemmas 1 and 2 that a shortest zookeeper's route is obtained when the current route becomes non-adjustable at all reflection points, including the starting point on the chosen cage. There are at most $O(n)$ event points in the sliding

process, and each of them can be maintained in linear time. See [8] for details (for simplicity, a brute-force algorithm was given in [8]).

Let us now briefly review the $\sqrt{2}$ -approximation algorithm for the fixed zookeeper's problem [11]. Let $P_1, P_2, \dots, P_m$ denote the cages indexed in clockwise scan of the boundary of the zoo P , starting at the point s . Let $s = s_0 = s_{m+1}$. First, we compute the point s_1 on the boundary of P_1 , which is closest to s_0 (in the geodesic distance) inside the interior of $P - \mathcal{P}$. Next, compute the point s_2 on the boundary of P_2 that is closest to s_1 , and so on. An important property of these computed points s_i is that they are to the left of the reflection points of the shortest zookeeper's route through s [11]. Putting together the shortest paths between s_i and s_{i+1} for all i , $0 \leq i \leq m$, gives a zookeeper's route, whose length is at most $\sqrt{2}$ times that of the shortest zookeeper's route through s . See Fig. 3(b) for an example.

3. The 2-approximation algorithm

The idea of our approximation algorithm is to select a point s on some cage of $\mathcal{P}$ and then compute the zookeeper's route through s using the known $\sqrt{2}$ -approximation algorithm [11]. The point s is chosen so that neither the incoming angle nor the outgoing angle of the shortest zookeeper's route through s with the edge containing s can exceed $\pi/2$. This enables us to obtain the approximation factor 2.

We will denote by $\overline{ab}$ the line segment having two endpoints a and b , and $|ab|$ the length of $\overline{ab}$. Also, denote by $|R|$ the Euclidean distance of a route R , and $\pi(x, y)$ the shortest path between two points x and y inside the interior of $P - \mathcal{P}$. We call the vertex of a cage P_i , which lies on the boundary of P and is first visited by a clockwise scan of P starting at a boundary point of P outside of P_i , the *left endpoint* of P_i , and the other vertex of P_i on P the *right endpoint* of P_i .

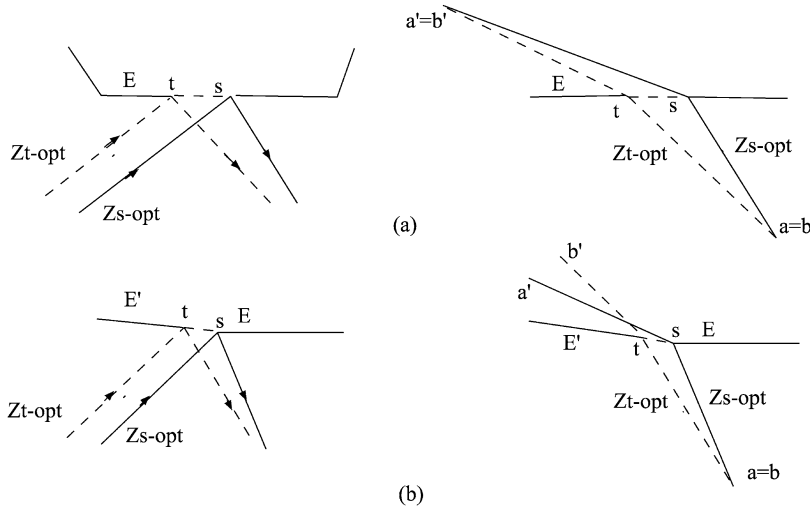


Fig. 4. Two situations for the unfolded routes Z_{s-opt} and Z_{t-opt} .

Let Z_{s-app} be the zookeeper's route through a point s , which is computed by the algorithm of [11]. Denote by Z_{s-opt} the shortest zookeeper's route through s , and Z_{opt} an overall shortest zookeeper's route. See Fig. 3 for an example. Then, we have the following main result of this paper.

Theorem 1. *For any instance of the zookeeper's route problem, we can find a point s on some cage of $\mathcal{P}$ such that $|Z_{s-app}| \leq 2|Z_{opt}|$ holds. Moreover, the route Z_{s-app} can be computed in $O(n)$ time.*

Proof. Assume that there are at least three cages inside P ; otherwise, a shortest zookeeper's route can simply be found in linear time. Let P_1 , P_2 and P_3 be three consecutive cages on the boundary of P in clockwise order. We now describe how to find the starting point s on the boundary of P_2 . First, compute two shortest paths between the right endpoint of P_2 and the left endpoint of P_1 and between the left endpoint of P_2 and the right endpoint of P_3 . Clearly, these two paths cross. Denote by p the intersection point of two paths closest to P_2 (in the geodesic distance). If p is a vertex of the polygon P , then any shortest zookeeper's route has to pass through p , and thus we let $s = p$. Otherwise, denote by s the point on the boundary of P_2 such that the length of $\pi(p, s)$ gives the closest distance between p and P_2 , and denote by E the edge of P_2 containing s . In this case, the path $\pi(p, s)$ consists of a single line segment $\overline{ps}$, and the angle at s between E and $\overline{ps}$ is $\pi/2$. See Fig. 3(a) for an example. For any shortest path that starts at a point on P_1 , then visits s and ends at a point on P_3 , neither the incoming angle nor the outgoing angle of

the shortest path with E can exceed $\pi/2$. Thus, neither the incoming angle nor the outgoing angle of the route Z_{s-opt} with E can exceed $\pi/2$.

It is easy to see that the point s described above can be found in $O(n)$ time. Next, we compute the route Z_{s-app} using the linear-time algorithm of [11]. Our main work is then to show that $|Z_{s-opt}| \leq \sqrt{2}|Z_{opt}|$, as the inequality $|Z_{s-app}| \leq \sqrt{2}|Z_{s-opt}|$ is already known [11]. Assume below that s is a point on some edge E of the cage P_2 ; otherwise, s is a vertex of P such that any shortest zookeeper's route passes through it and thus $|Z_{s-app}| \leq \sqrt{2}|Z_{opt}|$ holds.

Since Z_{s-opt} is the shortest zookeeper's route through s , it cannot be adjusted at all reflection points, except for the starting point s on the edge E . See Fig. 3(c) for an example. (If Z_{s-opt} cannot be adjusted at s either, then we have $Z_{s-opt} = Z_{opt}$ [8].) Let t denote the event point next to s on the chosen cage P_2 in the sliding process described in the previous section [8]. In the case that s is an endpoint of E , the point t is contained in a new edge E' that also has s as its endpoint (see Fig. 4(b)). Denote by Z_{t-opt} the shortest zookeeper's route through t . Then, we have $|Z_{t-opt}| < |Z_{s-opt}|$. See Fig. 4 for some examples, where solid and dotted lines show the routes Z_{s-opt} and Z_{t-opt} , respectively.

Denote by a an intersection point of Z_{s-opt} with Z_{t-opt} . Then, unfold the route Z_{s-opt} by taking a as the starting point. Let a' denote the point obtained by reflecting a across the last active edge. Since Z_{s-opt} is adjustable only on E at the point s , the unfolded route Z_{s-opt} can topologically be considered as two line segments $\overline{as}$ and $\overline{sa'}$. See Fig. 4. (In the case that the unfolded route Z_{s-opt} consists of more than two segments,

we can stretch out the unfolded route Z_{s-opt} along its two segments incident to s so that the stretched route consists of only two segments $\overline{as}$ and $\overline{sa'}$.) Since neither of the incoming angle nor the outgoing angle of Z_{s-opt} with E can exceed $\pi/2$, the angle $\angle asa'$ of the triangle $\triangle asa'$ is at least $\pi/2$. Thus, $|Z_{s-opt}| = |as| + |sa'| \leq \sqrt{2}|aa'|$ holds [11].

Let b be the starting point of Z_{t-opt} , which is identical to a , and b' the point obtained by reflecting b across the last active edge for the route Z_{t-opt} . Also, the unfolded route Z_{t-opt} can topologically be considered as two line segments $\overline{bt}$ and $\overline{tb'}$. See Fig. 4 for two different situations. Since $|Z_{t-opt}| < |Z_{s-opt}|$ holds, the distance from t to $\overline{bb'}$ is smaller than the distance from s to $\overline{aa'}$, so as to approach the route Z_{opt} whose unfolded version can be considered as a line segment. Furthermore, since the length function of the shortest zookeeper's routes through the points from s to t on the edge E or E' is monotonically decreasing, we have $|sa'| = |sb'|$. Hence, $|bb'| \geq |aa'|$ holds. See Fig. 4.

The route Z_{opt} can be obtained by sliding the reflection point of the current route on the boundary of P_2 in the adjusting direction, and computing the shortest zookeeper's routes through all encountered event points on P_2 , until the non-adjustable route Z_{opt} is obtained [9]. Since the unfolded route Z_{opt} can be considered as a line segment, we have $|aa'| \leq |bb'| \leq \dots \leq |Z_{opt}|$. Since $|Z_{s-opt}| \leq \sqrt{2}|aa'|$ holds, we obtain $|Z_{s-opt}| \leq \sqrt{2}|Z_{opt}|$. It completes the proof. $\square$

4. Conclusion

In this paper, we have proposed a simple approximation scheme for computing zookeeper's routes in simple polygons. Our approximation algorithm produces in $O(n)$ time a zookeeper's route of length at most 2 times that of the shortest zookeeper's route. Our result improves upon the previous approximation factor 6 [7].

It is worth to pointing out that the techniques we employ in this paper can probably be used for approx-

imating other kinds of routes as well. If we allow the considered route to enter the interior of a cage, then it introduces the *safari route problem* [10]. A more interesting problem, called the *watchman route problem*, is to find a shortest route from which every interior point of a given polygon is visible [9]. Our scheme can also be applied to these two problems to obtain the approximation factor 2. The remaining work is to develop the linear-time approximation algorithms for these problems. We are working in this direction.

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